

Clarifying Prompts for Robust English-Arabic Question-Type Classification

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Abstract—This paper studies prompt design for bilingual English-Arabic question-type classification. The task is to classify each question into one of six answer-type labels: NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, and ABBREVIATION. We conducted a controlled prompt progression study with Falcon3-7B-Instruct, moving from simple label-only prompts to prompts containing explicit label definitions, few-shot examples, and a targeted DESCRIPTION/ENTITY disambiguation rule. We then froze the best prompt strategy and evaluated it on a new balanced validation set with Falcon3, ALLaM, AceGPT, and Gemini-2.5-Flash. Each question was evaluated under eight prompt variants, and the final label was selected by majority vote. On the unseen validation set, Falcon3 reached 1.0000 English accuracy and 0.9833 Arabic accuracy; ALLaM reached 0.9833 English accuracy and 1.0000 Arabic accuracy; AceGPT reached 0.9167 English accuracy and 0.8833 Arabic accuracy; and Gemini reached 0.9833 English accuracy and 0.8333 Arabic accuracy. The main remaining closed-model failure mode was Arabic ABBREVIATION: under compact prompt variants, Gemini mapped all ten Arabic acronym-expansion questions to DESCRIPTION. Results suggest that definitions, examples, and targeted disambiguation can reduce prompt sensitivity in this controlled pilot setting, while larger and more natural benchmarks are needed before making broad claims about bilingual robustness.

Index Terms—prompt engineering, bilingual classification, Arabic NLP, question classification, LLM evaluation, prompt sensitivity, consistency

I. Introduction

Large language models (LLMs) are increasingly used as general-purpose interfaces for classification, question answering, tutoring, and information extraction. Their natural-language interface makes them flexible, but it also creates a methodological challenge: small changes in prompt wording can change predictions, especially in classifi-

cation tasks where label boundaries are semantically close [24], [18], [4]. Prompt engineering is therefore not only a practical skill, but also an experimental variable that should be controlled and evaluated.

This paper investigates bilingual prompt engineering for answer-type question classification in English and Arabic. The task is inspired by question classification settings such as TREC, where a question is assigned to the type of answer it requests [20], [12]. The six labels used here are NUMBER, LOCATION, PERSON, ENTITY, DESCRIPTION, and ABBREVIATION. Some labels have clear surface cues, while DESCRIPTION and ENTITY can overlap when a question asks what something is.

A further motivation for this study is the limited empirical evidence on recent Falcon-family instruction models in bilingual prompt-robust classification settings. Falcon3-7B-Instruct is a recent open Falcon-family instruction model, and its behavior in English-Arabic prompt-robust question classification remains underexplored [19], [28].

The study addresses three research questions. RQ1 asks how prompt design affects bilingual classification accuracy. RQ2 asks whether definitions, few-shot examples, and a targeted DESCRIPTION/ENTITY rule reduce sensitivity and improve consistency. RQ3 asks whether the frozen final prompt generalizes across open and closed instruction-tuned models on an unseen balanced validation set.

The contributions are threefold. First, we construct a small balanced English-Arabic validation set for six-way answer-type classification. Second, we compare prompt strategies from label-only prompting to targeted disambiguation. Third, we evaluate four models with accuracy, macro-

F1, sensitivity, consistency, class-level accuracy, confidence intervals, and error analysis. The study also contributes early empirical evidence on Falcon3-7B-Instruct in English-Arabic prompt-robust question classification, a setting not yet well covered by existing Falcon-family evaluations.

II. Related Work

A. Prompt Engineering and Few-Shot Learning

LLMs have shown strong few-shot learning ability when tasks are expressed through natural-language prompts [2], and instruction tuning made natural-language task following more reliable across downstream uses [16]. Prompt engineering research has explored demonstrations, task descriptions, formatting, reasoning instructions, output constraints, and self-consistency [22], [11], [17], [23], [21]. Prior work also shows that example order, label wording, and calibration can affect predictions [24], [14], [5].

B. Prompt Sensitivity, Consistency, and Robustness

A growing body of work shows that LLMs are sensitive to prompt wording and format. Sclar et al. quantify sensitivity to prompt formatting [18]. Errica et al. propose sensitivity and consistency as diagnostic metrics for classification tasks, where sensitivity measures changes across prompt rephrasings and consistency measures variation among examples of the same class [4]. Robustness work also studies semantically equivalent transformations, invariance, multi-call inference, autoregressive artifacts, and benchmark assumptions [13], [1], [3], [15].

C. Question Classification and Arabic Evaluation

Question classification was formalized in early question-answering pipelines as a way to predict the expected answer type before retrieval or answer extraction. The TREC question-answering test collection introduced broad answer categories that included numeric, location, person, description, entity, and abbreviation-like targets [20]. Li and Roth later showed that these answer-type categories could be learned as a supervised classification problem, making the taxonomy a standard reference point for question classification research [12]. Although the original setting predates modern instruction-tuned LLMs, the taxonomy remains useful in zero-shot and few-shot prompting because it gives models a compact and interpretable output space. Our label set therefore follows this tradition directly: NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, and ABBREVIATION.

Arabic adds linguistic challenges that make this taxonomy harder to apply without explicit prompt clarification. Arabic NLP is affected by rich morphology, clitic attachment, optional diacritics, orthographic variation, diglossia between Modern Standard Arabic and dialects, and right-to-left script processing [25]. In question classification, these properties can make surface cues less reliable than in English because a short Arabic question may combine interrogative particles, attached prepositions, and a target expression in a compact form. ABBREVIATION questions are especially vulnerable because Arabic formulations such as “what does the abbreviation X mean?” resemble definitional or meaning-seeking questions, which can push an LLM toward DESCRIPTION rather than ABBREVIATION. This linguistic ambiguity directly motivates our later error analysis, where Gemini-2.5-Flash classified all ten Arabic ABBREVIATION validation samples as DESCRIPTION.

Multilingual LLM evaluation also shows that model performance varies substantially by language, benchmark design, and model family [6], [7]. For this reason, English results cannot be assumed to transfer directly to Arabic, even when the task labels and prompt format are held constant. We include AceGPT and ALLaM because both are Arabic-specialized or Arabic-focused instruction models, making them important comparisons against a general open model and a closed Gemini model [8], [9]. Falcon-family models have been widely discussed as open foundation models, but Falcon3-specific empirical studies remain comparatively limited, especially for Arabic-facing prompt sensitivity and answer-type classification [28], [19]. This paper therefore uses Falcon3-7B-Instruct not only as an experimental model, but also as a case study for evaluating a recent open model against Arabic-focused models such as ALLaM and AceGPT. Broader Arabic evaluation efforts such as ALUE and ArabicMMLU show that Arabic NLU evaluation increasingly requires dedicated benchmarks rather than translated English-only test sets [26], [27]. Our validation set is smaller and task-specific, but it follows the same motivation by evaluating bilingual English-Arabic behavior under controlled label definitions before the methodology section defines the task and dataset construction.

TABLE I
Validation Dataset Distribution

Language	NUM.	LOC.	PER.	DESC.	ENT.	ABBR.	Total
English	10	10	10	10	10	10	60
Arabic	10	10	10	10	10	10	60
Total	20	20	20	20	20	20	120

III. Methodology and Benchmark Construction

A. Task Definition

The task is single-label classification. Given a question x , the model outputs one label y from

$$Y = \{\text{NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, ABBREVIATION}\}. \quad (1)$$

NUMBER covers numbers, dates, quantities, calculations, or measurements. LOCATION covers places and geographical areas. PERSON covers named humans or human roles. DESCRIPTION covers explanations, meanings, definitions, functions, uses, or processes. ENTITY covers specific non-person, non-location, non-number things. ABBREVIATION covers acronyms and their expansions.

B. Dataset Construction

The final validation dataset contains 120 questions: 60 English and 60 Arabic. Each language contains 10 examples for each of the six labels. The dataset distribution is shown in Table I.

The dataset was developed through iterative pilots. Early experiments exposed ambiguity in ENTITY, especially when examples overlapped with LOCATION, ABBREVIATION, or DESCRIPTION. The final validation set was created as an unseen balanced dataset after the main prompt-refinement stage.

Although this validation set is smaller than benchmark-scale datasets, it was designed as a controlled pilot-scale prompt-engineering microbenchmark rather than as a general-purpose evaluation benchmark. The balanced design made every language-label pair manually inspectable, kept the six classes equally represented, and enabled direct class-level comparison between English and Arabic. Because each item was evaluated under eight prompt variants and across four models, the final unseen cross-model validation contains 3,840 prompt-level generations. This repeated-measures design supports diagnostic analysis of prompt sensitivity, consistency, and majority-vote stability, while the dataset size limits broad generalization.

C. Prompt Strategy and Majority Voting

Pilot 03 used simple label-only prompts. Pilot 04 added explicit definitions. Pilot 05 added one few-shot example per label. Pilot 06 retained definitions and examples and added a targeted DESCRIPTION/ENTITY rule. Pilot 07 reused the exact Pilot 06 strategy on unseen validation data. Gemini was evaluated with eight compact variants of the same frozen final strategy to keep the closed-model prompts short while preserving the same label definitions and disambiguation rule. For Gemini-2.5-Flash, the same 120-item unseen validation set was evaluated under eight compact prompt variants, producing 960 closed-model generations in total: 480 English generations and 480 Arabic generations. Therefore, the Gemini result reflects majority-vote behavior across prompt variants rather than a single-prompt evaluation.

Because the eight variants were task-instruction variants rather than subjective politeness levels, no native-speaker politeness questionnaire was used; prompt validity was instead controlled by preserving the same label set, label definitions, examples, disambiguation rule, and exact-label-only output constraint across variants.

For each question x_i , the final prediction $\hat{y}_i$ was selected by majority vote:

$$\hat{y}_i = \arg \max_{c \in Y} \sum_{j=1}^8 \mathbf{1}[f(x_i, p_j) = c]. \quad (2)$$

D. Evaluation Metrics

The evaluation used accuracy, macro-F1, average sensitivity, average consistency, class-level accuracy, error analysis, bootstrap confidence intervals, and paired sign-test p-values. Paired tests compare sample-level correctness vectors between two models on the same validation examples. We used paired sign tests, equivalently exact binomial tests on discordant pairs, because the same examples were evaluated by each model. Since each language subset contains only 60 validation examples and several models are near-perfect, the resulting p-values are descriptive rather than definitive. Sensitivity is normalized entropy over the prediction distribution $q_i(c)$ across prompt-level outputs:

$$S(x_i) = -\frac{\sum_{c \in Y} q_i(c) \log q_i(c)}{\log |Y|}. \quad (3)$$

Consistency between same-class examples follows total-variation-distance similarity:

$$C(x_i, x_k) = 1 - \frac{1}{2} \sum_{c \in Y} |q_i(c) - q_k(c)|. \quad (4)$$

TABLE II
Reproducibility details for deterministic reruns where supported.

Item	Value
Environment	Colab and local Python 3.x
HF GPU	Tesla T4 GPU
Temperature	0.0
Random seed	42 where applicable
HF quantization	4-bit NF4
Gemini model	gemini-2.5-flash
Gemini max output tokens	64
Gemini thinking budget	0
Gemini generations	960 total: 480 English, 480 Arabic
Prompt variants	8 per language / 8 per sample
Majority voting	Yes

IV. Experimental Settings

A. Models

The tested systems were Falcon3, ALLaM, AceGPT, and Gemini [19], [9], [8], [10]. Falcon3 was used during iterative prompt design. The final prompt was then frozen and evaluated on all four models. The exact model identifiers were tiiaue/Falcon3-7B-Instruct, humain-ai/ALLaM-7B-Instruct-preview, FreedomIntelligence/AceGPT-v2-8B-Chat, and gemini-2.5-flash.

B. Runtime Environment and Reproducibility

Experiments were conducted in Google Colab and local Python for the Gemini rerun. Hugging Face models were loaded with 4-bit NF4 quantization using Transformers and BitsAndBytes. Gemini was called through the Google GenAI client. Table II summarizes the reproducibility settings.

Across the four final models, the unseen validation stage produced 3,840 prompt-level generations. This total reflects 120 validation questions evaluated under eight prompt variants for each of the four models.

For reproducibility, the repository includes raw CSV outputs, prompt files, result summaries, and scripts for regenerating figures and tables from the saved outputs. Full model inference may require GPU access for the Hugging Face models and a Gemini API key for the closed model. API keys should be supplied through environment variables or notebook secrets and must not be committed to the repository.

C. Pilot Progression

Table III summarizes the Falcon3 prompt progression from Pilot 03 to Pilot 07. The progression was not used to tune the unseen validation set; it was used to select a final prompt strategy before cross-model validation.

TABLE III
Falcon3 prompt progression. Pilot 07 used the frozen final strategy on unseen balanced validation data.

Pilot	Prompt design	Eng. Acc.	Ar. Acc.
Pilot 03	Simple labels	0.8667	0.6667
Pilot 04	Definitions	0.9500	0.8000
Pilot 05	Definitions + few-shot	0.9667	0.8667
Pilot 06	Few-shot + DESC./ENT. rule	0.9833	0.9500
Pilot 07	Frozen prompt, unseen data	1.0000	0.9833

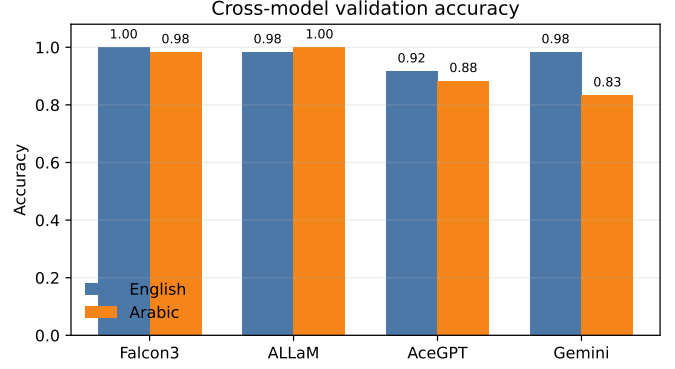


Fig. 1. Cross-model majority-vote accuracy by language on the unseen balanced validation set. Higher bars indicate stronger aggregate performance.

V. Results

A. Cross-Model Validation

Table IV shows that the final prompt generalized across the tested models, but not uniformly. Falcon3 and ALLaM reached near-perfect validation accuracy in both languages. AceGPT remained strong but showed more concentrated ENTITY weaknesses. Gemini performed strongly in English but lower in Arabic because the compact prompt variants consistently treated Arabic acronym-expansion questions as DESCRIPTION rather than ABBREVIATION. In Table IV, Falcon3, ALLaM, AceGPT, and Gemini refer respectively to tiiaue/Falcon3-7B-Instruct, humain-ai/ALLaM-7B-Instruct-preview, FreedomIntelligence/AceGPT-v2-8B-Chat, and gemini-2.5-flash. The Gemini values in Table IV therefore summarize majority-vote behavior over 960 closed-model generations rather than single-prompt predictions.

B. Class-Level Results

Class-level accuracy is reported in Table V and visualized in Fig. 4. The most visible differences are AceGPT’s lower ENTITY accuracy and Gemini’s Arabic ABBREVIATION failure.

C. Statistical Tests and Confidence Intervals

Table VI reports bootstrap 95% confidence intervals for validation accuracy. Fig. 5 and Fig. 6

TABLE IV
Cross-model validation on the unseen balanced dataset.

Model	Eng. Acc.	Ar. Acc.	Eng. F1	Ar. F1	Eng. Sens.	Ar. Sens.	Eng. Cons.	Ar. Cons.
Falcon3	1.0000	0.9833	1.0000	0.9833	0.0048	0.0105	0.9917	0.9542
ALLaM	0.9833	1.0000	0.9833	1.0000	0.0059	0.0059	0.9833	0.9833
AceGPT	0.9167	0.8833	0.9087	0.8618	0.0385	0.0313	0.8787	0.8824
Gemini	0.9833	0.8333	0.9833	0.7778	0.0253	0.0225	0.9662	0.9704

TABLE V
Class-level majority-vote accuracy. Each cell summarizes 10 validation examples.

Language	Label	Falcon3	ALLaM	AceGPT	Gemini
English	NUMBER	1.0	1.0	1.0	1.0
English	LOCATION	1.0	1.0	1.0	1.0
English	PERSON	1.0	1.0	1.0	1.0
English	DESCRIPTION	1.0	1.0	1.0	1.0
English	ENTITY	1.0	0.9	0.5	1.0
English	ABBREVIATION	1.0	1.0	1.0	0.9
Arabic	NUMBER	1.0	1.0	1.0	1.0
Arabic	LOCATION	1.0	1.0	1.0	1.0
Arabic	PERSON	0.9	1.0	1.0	1.0
Arabic	DESCRIPTION	1.0	1.0	1.0	1.0
Arabic	ENTITY	1.0	1.0	0.3	1.0
Arabic	ABBREVIATION	1.0	1.0	1.0	0.0

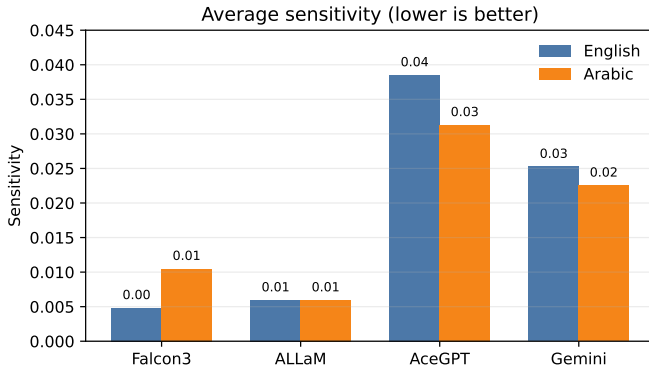


Fig. 2. Average prompt sensitivity by model and language across eight variants. Lower values indicate less variation across prompt phrasings, but low sensitivity does not guarantee correctness.

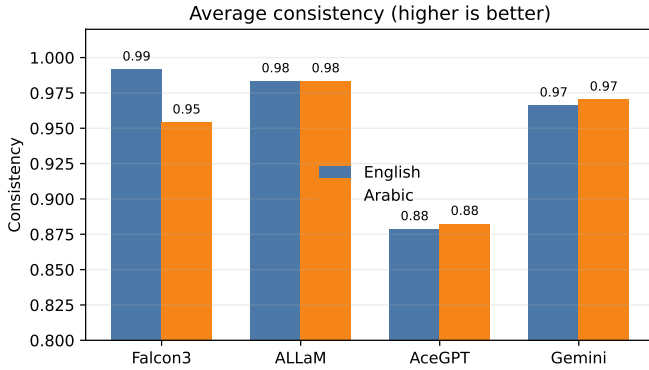


Fig. 3. Average within-class consistency by model and language. Higher values indicate more similar prediction distributions among examples with the same gold label.

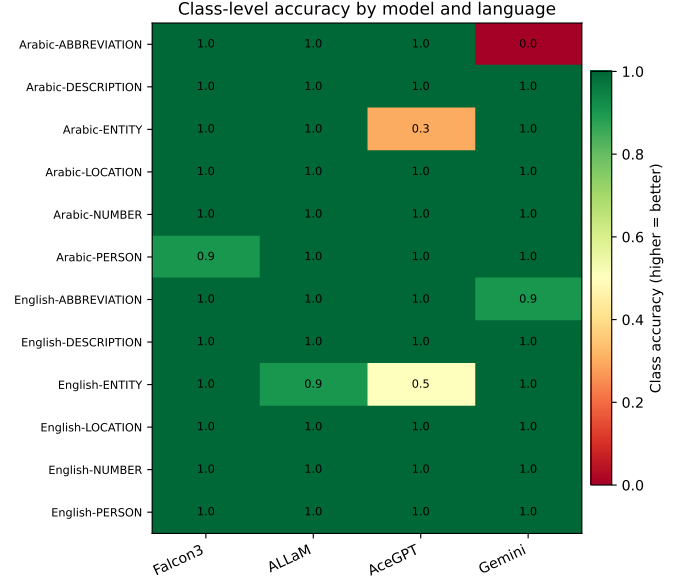


Fig. 4. Class-level majority-vote accuracy heatmap. Cell text and color both encode accuracy for 10 examples per language-label pair; higher-scale cells indicate higher class-level accuracy, while lower-scale cells indicate lower class-level accuracy. Gemini’s Arabic ABBREVIATION failure is visible as the lowest cell.

summarize paired sign-test p-values computed from paired sample-level correctness vectors. The confidence intervals are wide for non-perfect scores because the validation set contains 60 examples per language. Paired tests compare sample-level correctness vectors between two models on the same validation examples; we used paired sign tests, equivalently exact bino-

TABLE VI
Bootstrap 95% confidence intervals for validation accuracy.

Model	Lang.	Acc.	CI Low	CI High
Falcon3	English	1.0000	1.0000	1.0000
Falcon3	Arabic	0.9833	0.9500	1.0000
ALLaM	English	0.9833	0.9500	1.0000
ALLaM	Arabic	1.0000	1.0000	1.0000
AceGPT	English	0.9167	0.8500	0.9667
AceGPT	Arabic	0.8833	0.8000	0.9500
Gemini	English	0.9833	0.9500	1.0000
Gemini	Arabic	0.8333	0.7333	0.9167

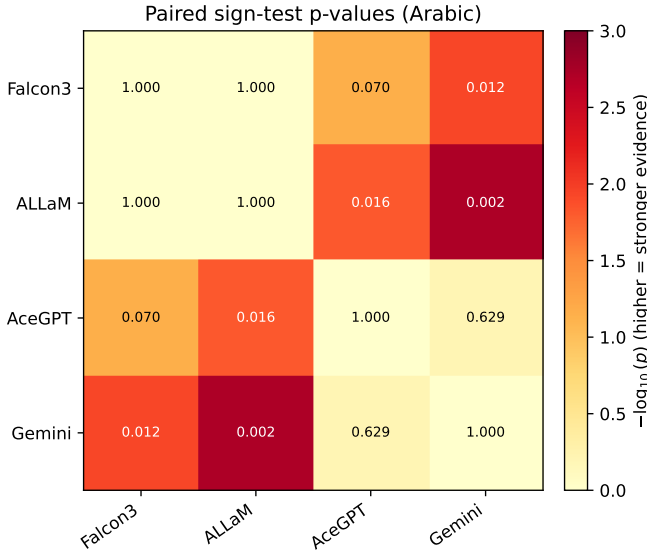


Fig. 5. Arabic paired sign-test heatmap. Cell text reports p -values and color encodes $-\log_{10}(p)$; darker cells indicate stronger evidence against equal paired accuracy. Because the Arabic subset has 60 examples and several models are near-perfect, these results should be interpreted cautiously.

mial tests on discordant pairs, because the same examples were evaluated by each model. Since each language subset contains only 60 validation examples and several models are near-perfect, the resulting p -values are descriptive rather than definitive. The reproducibility repository includes a statistical-test script for regenerating the p -value tables and heatmaps.

VI. Discussion

In this controlled pilot setting, the Falcon3 prompt progression supports RQ1 and RQ2. Moving from label-only prompting to explicit definitions improved label-boundary clarity. Adding few-shot examples stabilized Arabic predictions, and the targeted DESCRIPTION/ENTITY rule fixed earlier cases where Arabic DESCRIPTION-style cues were confused with ENTITY. The results suggest that part of the apparent bilingual

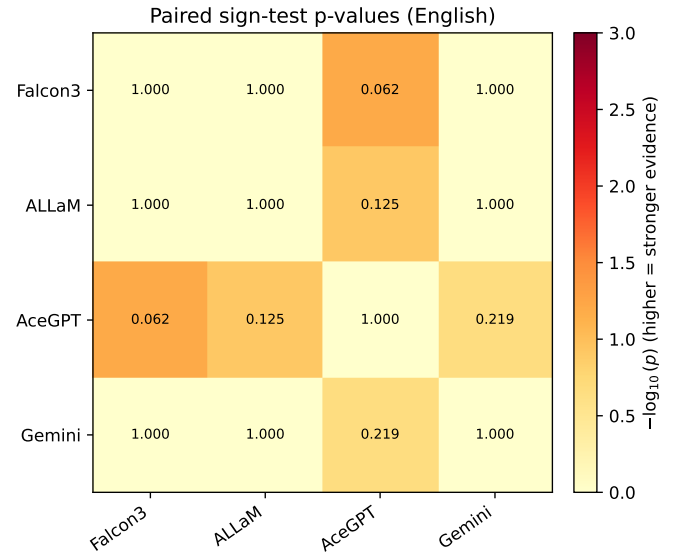


Fig. 6. English paired sign-test heatmap. Cell text reports p -values and color encodes $-\log_{10}(p)$; darker cells indicate stronger evidence against equal paired accuracy. The small validation size and near-perfect scores make the p -values descriptive rather than definitive.

difficulty came from underspecified task boundaries rather than from the language alone.

RQ3 is partly supported. The final prompt generalized across the tested models, but not uniformly: Falcon3 and ALLaM were near-perfect, AceGPT remained strong while struggling mainly with ENTITY, and Gemini performed strongly in English but exposed an Arabic ABBREVIATION failure under compact prompt variants. Arabic questions phrased as “What does the abbreviation X mean?” were interpreted as DESCRIPTION rather than ABBREVIATION. This failure is analytically important because low sensitivity does not guarantee correctness when a model is stably wrong.

The value of the dataset is therefore diagnostic rather than exhaustive. Its balanced structure allows controlled comparison of prompt strategies and model-specific label-boundary failures, including AceGPT’s ENTITY weakness and Gemini’s Arabic ABBREVIATION failure. These patterns should be treated as hypotheses for larger studies rather than as definitive population-level conclusions.

Accuracy, sensitivity, consistency, and error analysis must therefore be interpreted together. Low sensitivity is useful only when the stable behavior is correct, while high aggregate accuracy can hide label-specific failures. Further validation is needed on larger, more natural, and more linguistically diverse English-Arabic question sets.

VII. Conclusion

This study evaluated prompt design for English-Arabic question-type classification in a controlled pilot setting. The best prompt combined definitions, few-shot examples, and a targeted DESCRIPTION/ENTITY disambiguation rule. On unseen validation, this design reached near-perfect results for Falcon3 and ALLaM and strong results for AceGPT. Gemini 2.5 Flash achieved 0.9833 English accuracy and 0.8333 Arabic accuracy under eight compact variants, with Arabic ABBREVIATION as the dominant failure mode. The main conclusion is that targeted prompt clarification can reduce bilingual label confusion in this dataset, while cross-model behavior remains uneven. Further validation on larger and more natural datasets is needed.

Limitations

The dataset size is the main limitation of the study. The validation set is small, balanced, and manually constructed: it contains 120 validation examples, with only 10 examples per class per language. The results should therefore be interpreted as evidence from a controlled prompt-engineering microbenchmark, not as a definitive benchmark for Arabic NLP or multilingual question classification. The dataset is not representative of natural question distributions, dialectal Arabic was not evaluated, and the topics are limited. The prompt was refined through prior error analysis, so the reported results should be read as controlled validation rather than as a broad benchmark. Gemini compact variants may have reduced abbreviation guidance. Hugging Face models were run with 4-bit quantization, which can differ from full-precision inference, and API models can change over time. No questionnaire was used because the prompt variants are task-instruction variants rather than subjective politeness variants. Larger datasets with natural user questions, dialectal Arabic, and more varied topics are needed.

Ethics Statement

This project uses a small educational dataset and does not require private or sensitive user data. The main ethical risk is overclaiming from a small benchmark. To reduce this risk, we report raw predictions, prompts, metrics, confidence intervals, and error analysis. API keys are not stored in the repository and must be supplied through environment variables or notebook secrets. Multilingual evaluation should avoid treating Arabic as inherently deficient; observed failures are model-, prompt-, and dataset-specific.

Acknowledgements

This project was completed for the Prompt Engineering course at the Mediterranean Institute of Technology. We thank Professor Abdeldjalil Labeled for the project guidelines and evaluation framework. We also acknowledge the open-source model developers, the Hugging Face ecosystem, and the Google Gemini API ecosystem.

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Appendix A

Prompt and Reproducibility Summary

The final frozen prompt strategy used explicit label definitions, one example per label, a DESCRIPTION/ENTITY disambiguation rule, and an output constraint requiring exactly one label. Eight variants were used per language. Full prompts, raw outputs, metrics, and figure-generation files are available in the repository.

The label definitions were: NUMBER for numbers, dates, quantities, calculations, or measurements; LOCATION for places and geographical areas; PERSON for named humans or human roles; DESCRIPTION for explanations, meanings, definitions, functions, uses, or processes; ENTITY for specific non-person, non-location, non-number things; and ABBREVIATION for acronyms and their expansions.

The key disambiguation rule was: choose DESCRIPTION when the question asks for an explanation, meaning, definition, function, purpose, use, or process; choose ENTITY when it asks for a specific object, animal, planet, language, currency, device, material, gas, software, element, metal, or instrument. The output constraint required exactly one of the six English label names and no extra text.

Listing 1. Representative English prompt variant.

```
Classify the question into exactly one label:
NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, ABBREVIATION
.

Definitions: NUMBER = number/date/quantity;
LOCATION = place; PERSON = human;
DESCRIPTION = explanation/meaning/function/process;
ENTITY = specific non-person/non-location thing;
ABBREVIATION = acronym or expansion.

Rule: DESCRIPTION for explanation, meaning,
definition, function, use, or process.
ENTITY for a specific thing.

Question: {question}
Output exactly one label only.
```

To avoid corrupted Arabic rendering in the final PDF, the representative Arabic compact variant is described in English rather than printed as Arabic script: insert the Arabic question into the same question field, preserve the six English label names and definitions, keep the DESCRIPTION/ENTITY rule, and require a single exact English label as output. For example, Arabic acronym-expansion questions phrased as “What does the abbreviation X mean?” should be labeled ABBREVIATION rather than DESCRIPTION.

Appendix B

Statistics Code Summary

The statistical analysis used saved sample-level correctness vectors. The repository includes a statistical-test script for regenerating paired p-value CSVs, the English p-value heatmap, the Arabic p-value heatmap, and bootstrap confidence intervals when the corresponding raw CSV files are present.

Listing 2. Bootstrap accuracy interval pseudocode.

```
def bootstrap_accuracy_ci(correct, n_boot=10000, seed=42):
    rng = np.random.default_rng(seed)
    samples = resample(correct, n_boot, rng)
    scores = [mean(sample) for sample in samples]
    return percentile(scores, [2.5, 97.5])
```

For each model pair and language, the paired sign test compared two correctness vectors on the same 60 examples. Let b be the number of examples correct for model A and wrong for model B, and let c be the number correct for model B and wrong for model A. Tied examples were ignored, and the exact binomial test was applied to the $b + c$ discordant pairs under equal probability. These tests are descriptive because the validation subsets are small and several models have near-perfect scores.